Project: Worked examples
Title: Plane wall heat transfer (layers and films)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the steady heat flow through a composite plane wall, between two fluids or between two known surface temperatures. It gives the overall U-value, the heat flux and every surface and interface temperature.

WHAT IT CALCULATES	RELATION
Resistance of each surface film	$R = 1/(h \cdot A)$
Resistance of each conducting layer	$R_i = t_i/(k_i \cdot A)$
Total resistance, hot side first	$R_{tot} = \Sigma R$
Heat flow through the wall	$\Phi = (T_1 - T_2)/R_{tot}$
Heat flux and overall U-value	$q = \Phi/A$; $U_{wall} = 1/(R_{tot} \cdot A)$
Surface and interface temperatures, stepping down from T_1 across each resistance in turn	$T_{next} = T_{previous} - \Phi \cdot R$

Basis: The plane wall: thermal resistance and the composite wall[1](§3.1, 7th ed.); Generalized thermal resistance networks (multilayer plane walls)[2](§3-3, 5th ed.); Thermal resistance of building components (method)[3](cl. 6).

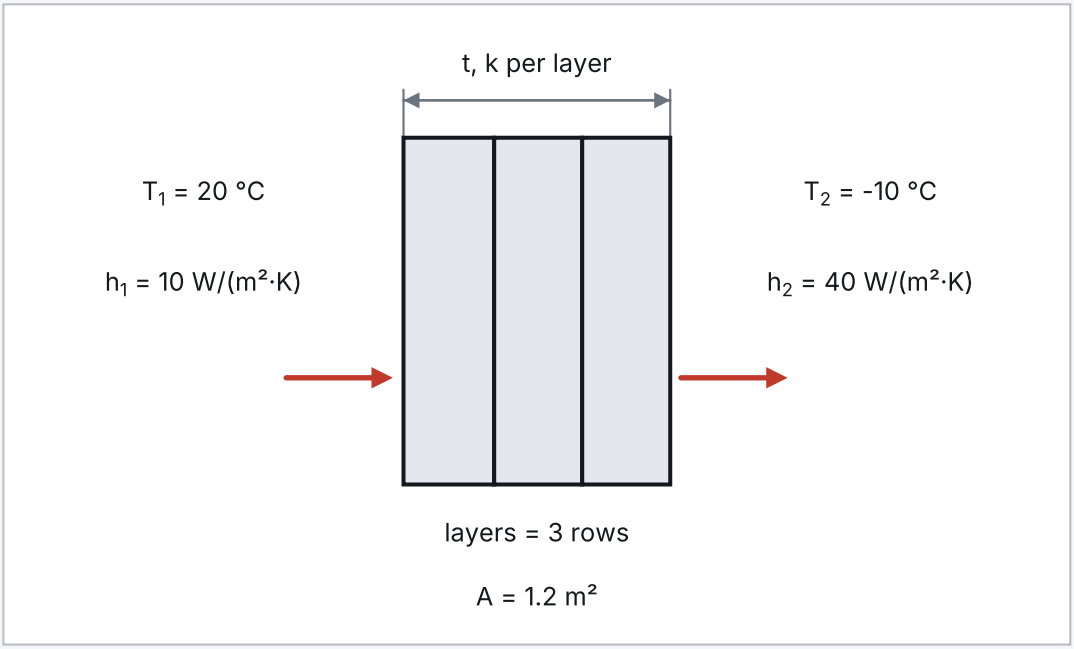


Figure 1 — A composite plane wall, side 1 left: a fluid film, the layers as listed (three drawn), a film on side 2. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers	Conducting layers, side 1 first	3 row(s)		drawing: DRG-GL-802 rev A	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Wall area normal to the heat flow	1.2	m²	drawing: DRG-GL-802 rev A	0.001 … 10000 m²
T_1	Side 1 temperature (the fluid's, or the surface's when no film is given)	20	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C
T_2	Side 2 temperature (the fluid's, or the surface's when no film is given)	-10	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C
h_1	Side 1 film coefficient (optional; blank = no film, so T_1 is the surface temperature)	10	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
h_2	Side 2 film coefficient (optional; blank = no film, so T_2 is the surface temperature)	40	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)

TABLE ROW	VALUES
layers[1]	name = glass; t = 4 mm; k = 0.78 W/(m·K)
layers[2]	name = air gap; t = 10 mm; k = 0.026 W/(m·K)
layers[3]	name = glass; t = 4 mm; k = 0.78 W/(m·K)

3. Assumptions

A1. Steady, one-dimensional conduction through plane layers of constant conductivity, with no internal heat generation. (Thermal-circuit idealisation: equal layer areas, no edge losses.)
A2. Layers are in perfect thermal contact. (A contact resistance can be entered as a thin layer of equivalent thickness and conductivity.)
A3. Each film coefficient is constant over its surface. (Thermal-circuit idealisation.)

4. Working

[6] $R_{h1} = 1 / (h_1 \cdot A)$ [1](§3.1)
$= 1 / (10\ \mathrm{W/(m^2 \cdot K)} \times 1.2\ \mathrm{m^2})$
$= 0.083333\ \mathrm{K/W}$
Side 1 film resistance

[7] $R_1 = t_1 / (k_1 \cdot A)$ [1](§3.1)
$= 4\ \mathrm{mm} / (0.78\ \mathrm{W/(m \cdot K)} \times 1.2\ \mathrm{m^2})$
$= 0.0042735\ \mathrm{K/W}$
Conduction resistance of glass

[8] $R_2 = t_2 / (k_2 \cdot A)$ [1](§3.1)
$= 10\ \mathrm{mm} / (0.026\ \mathrm{W/(m \cdot K)} \times 1.2\ \mathrm{m^2})$
$= 0.32051\ \mathrm{K/W}$
Conduction resistance of air gap

[9] $R_3 = t_3 / (k_3 \cdot A)$ [1](§3.1)
$= 4\ \mathrm{mm} / (0.78\ \mathrm{W/(m \cdot K)} \times 1.2\ \mathrm{m^2})$
$= 0.0042735\ \mathrm{K/W}$
Conduction resistance of glass

[10] $R_{h2} = 1 / (h_2 \cdot A)$ [1](§3.1)
$= 1 / (40\ W/(m^2 \cdot K) \times 1.2\ m^2)$
$= 0.020833\ K/W$
Side 2 film resistance
[11] $R_{tot} = \Sigma R$ [2](§3-3)
$= 0.083333 + 0.0042735 + 0.32051 + 0.0042735 + 0.020833\ K/W$
$= 0.43323\ K/W$
Total thermal resistance
[12] $\Phi = (T_1 - T_2) / R_{tot}$
$= (20\ degC - (-10\ degC)) / 0.43323\ K/W$
$= 0.069248\ kW$
Heat flow from side 1 to side 2
[13] $q = \Phi / A$
$= 0.069248\ kW / 1.2\ m^2$
$= 57.707\ W/m^2$
Heat flux
[14] $U_{wall} = 1 / (R_{tot} * A)$
$= 1 / (0.43323\ K/W \times 1.2\ m^2)$
$= 1.9236\ W/(m^2 \cdot K)$
Overall heat transfer coefficient
[15] $T_{s0} = T_1 - \Phi \cdot R_{h1}$
$= 20\ °C - 69.248\ W \times 0.083333\ K/W$
$= 14.229\ degC$
Side 1 surface
[16] $T_{s1} = T_{s0} - \Phi \cdot R_1$
$= 14.229\ °C - 69.248\ W \times 0.0042735\ K/W$
$= 13.933\ degC$
Interface 1
[17] $T_{s2} = T_{s1} - \Phi \cdot R_2$
$= 13.933\ °C - 69.248\ W \times 0.32051\ K/W$
$= -8.2614\ degC$
Interface 2
[18] $T_{s3} = T_{s2} - \Phi \cdot R_3$
$= -8.2614\ °C - 69.248\ W \times 0.0042735\ K/W$
$= -8.5573\ degC$
Side 2 surface

5. Results

SYMBOL	VALUE	DESCRIPTION
Φ	0.06925 kW	Heat flow
q	57.71 W/m²	Heat flux
U_{wall}	1.924 W/(m²·K)	Overall heat transfer coefficient
R_{tot}	0.4332 K/W	Total thermal resistance
R_{h1}	0.08333 K/W	Side 1 film resistance
R_1	0.004274 K/W	Layer 1 resistance
R_2	0.3205 K/W	Layer 2 resistance
R_3	0.004274 K/W	Layer 3 resistance
R_{h2}	0.02083 K/W	Side 2 film resistance
T_{s0}	14.23 °C	Side 1 surface temperature

SYMBOL	VALUE	DESCRIPTION
T_{s1}	13.93 °C	Interface 1 temperature
T_{s2}	-8.261 °C	Interface 2 temperature
T_{s3}	-8.557 °C	Side 2 surface temperature

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

`Template bs.plane-wall@1.1.0 Content hash 3d94130d2f7a7b86 Engine calc-engine/1.0.0`
`Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
2. Çengel YA, Ghajar AJ. Heat and Mass Transfer: Fundamentals and Applications. 5th ed. McGraw-Hill Education; 2015.
3. British Standards Institution. Building Components and Building Elements—Thermal Resistance and Thermal Transmittance—Calculation Methods. BS EN ISO 6946. BSI.